

SAMADHI KANDEWELA

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Professional Summary

Final year Computer Engineering undergraduate specializing in Software Engineering and Machine Learning, with hands-on experience in backend development, scalable microservices and AI driven applications. Proficient in Java, Spring Boot, Python, SQL, and cloud technologies, with experience building full stack systems and machine learning solutions. Passionate about designing secure, maintainable and scalable software systems.

Education

B.Sc. Eng. (Hons) in Computer Engineering **2022 – Present**

University of Sri Jayewardenepura, Faculty of Engineering, Sri Lanka

- Current GPA: 3.50 / 4.00
- Minor Specialization: Data Management

Institute of Java Software Engineering (IJSE), Galle **2021**

Southlands College, Galle **2006 – 2019**

- G.C.E. Advanced Level: Combined Mathematics – A, Physics – B, Chemistry – A

Work Experience

Software Engineer Intern March 2025 – September 2025

Quantum Tunnel Design Pvt Ltd *Sri Lanka*

- Developed responsive web applications using React, HTML and CSS with reusable and maintainable UI components.
- Designed and integrated RESTful APIs with backend services for scalable web applications.
- Collaborated in Agile teams using Git based workflows, participating in code reviews and debugging.
- Improved software quality through testing, debugging and clean coding practices while contributing to maintainable solutions.

Projects

Automated Coronary Angiogram Analysis System Sept 2025 – Jul 2026

- Developed a real-time AI system for coronary angiogram analysis to detect, localize and assess coronary artery stenosis.
- Designed an end-to-end pipeline combining vessel segmentation, stenosis detection, QCA-based measurements, Grad-CAM explainability and 3D coronary visualization.
- Conducted clinical validation planning using Dice, IoU, precision, recall, F1 score and latency metrics.
- **Tech Stack:** Python, PyTorch, TensorFlow, OpenCV, YOLO, U-Net, Grad-CAM, QCA.
- **GitHub:** github.com/Samadhi-Kandewela/angio-ai

HeartSync – Microservices Based Decision Support Platform March 2026 – Jun 2026

- Designed and developed a scalable microservices based backend using Spring Boot and Spring Cloud.
- Implemented API Gateway, Eureka Service Discovery, JWT authentication and RBAC for secure communication.
- Integrated RabbitMQ for asynchronous event driven communication between services.
- Containerized microservices using Docker to support scalable and portable deployments.
- **Tech Stack:** Java, Spring Boot, Spring Cloud Gateway, React, PostgreSQL, RabbitMQ, Docker.
- **GitHub:** github.com/UshaniSanjana/heart-sync

TrendScope – YouTube Trend Predictor

Jan 2025 – Jan 2026

- Developed RESTful backend APIs using Flask to process YouTube and Kaggle data for trend analysis.
- Designed and integrated a PostgreSQL database for efficient storage and retrieval of analytics data.
- Built a full stack analytics platform by integrating backend services with a React frontend.
- Implemented machine learning models for NLP preprocessing, topic clustering, viral trend prediction and AI assisted content strategy generation.
- **Tech Stack:** Python, Flask, PostgreSQL, React, TypeScript, Tailwind CSS, scikit-learn, Pandas, TF-IDF, K-Means, Random Forest.
- **GitHub:** Frontend | Backend

SpotFix – Faculty Issue Reporting and Tracking System

Feb 2026

- Built a campus issue reporting platform with photo uploads, location details, status tracking and role based dashboards.
- Developed backend integrated role-based dashboards and implemented PostgreSQL data management with secure Row Level Security.
- **Tech Stack:** React, TypeScript, Tailwind CSS, shadcn/ui, PostgreSQL, Row Level Security.

Missing Person Finder System – Disaster Response Platform

Dec 2025

- Developed and launched a web based missing person finder platform during the Sri Lanka flood disaster to support rapid public reporting and volunteer based verification.
- Implemented missing person submission workflows and data management and deployed the application using Netlify and DirectAdmin to provide reliable access for affected families, student volunteers and coordinators.
- **Tech Stack:** React, TypeScript, Supabase, Netlify, DirectAdmin, GitHub.
- **GitHub:** <https://github.com/DahamAdikari/missing-person-disaster-management.git>

Latency-Aware Coronary Artery Segmentation

Sept 2025 – Jan 2026

- Designed SLIM-Net, a real time coronary vessel segmentation model using MobileNetV3-Large, depthwise separable decoder blocks and CBAM attention.
- Applied CLAHE preprocessing and elastic augmentation, achieving a validation Dice score of 0.902 with 1.81 GFLOPs.
- Benchmarked against U-Net, DeepLabV3+, SegFormer-B4 and ResU-Net using Dice, IoU, latency, FPS, FLOPs and parameter count.
- **Tech Stack:** Python, PyTorch, MobileNetV3, CBAM, OpenCV, CUDA.
- **GitHub:** <https://github.com/Samadhi-Kandewela/coronary-vessel-segmentation>

Human Following Robot – Computer Vision & Image Processing

Jan 2026

- Developed a robotic vision system capable of tracking and following a person in dynamic environments.
- Implemented YOLOv8 based person detection, IoU based target locking, MediaPipe pose estimation and multi layer distance filtering.
- **Tech Stack:** YOLOv8, MediaPipe, Python, OpenCV, Depth Sensors.
- **GitHub:** <https://github.com/UshaniSanjana/TurtleBot>

HealHub – Mental Health Support Application

Oct 2023 – Jan 2024

- Developed a mobile app connecting users with mental health professionals, resources, emergency hotlines and AI chatbot support.
- Implemented meeting scheduling, real-time chat, Firebase backend services and Google Meet API integration.
- **Tech Stack:** Flutter, Dart, Firebase, Dialogflow, Google Meet API, GitHub, Figma.
- **GitHub:** <https://github.com/shafeekms10/healhub.git>

RoofMate – Boarding House Finder Application

March 2024 – Sept 2024

- Developed a mobile app for finding, filtering, booking and managing boarding house listings.

- Integrated real time chat, favorites, Firebase backend services and Google Maps API for navigation.
- **Tech Stack:** Flutter, Dart, Firebase, Google Maps API, GitHub, Figma.
- **GitHub:** <https://github.com/raveens9/roofmate>

Technical Skills

Programming Languages: Java, Python, C++, JavaScript, TypeScript, SQL, Dart

Backend Technologies: Spring Boot, Spring Cloud, Flask, REST APIs, SQLAlchemy

Frontend: React, Flutter, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, Firebase, SQLite

Cloud & DevOps: Docker, Git, GitHub, AWS Lambda, AWS Educate

Machine Learning: TensorFlow, PyTorch, OpenCV, scikit-learn

Certifications

AWS Educate – Introduction to Cloud 101

Cisco Networking Academy – Introduction to Cybersecurity

Meta/Coursera – HTML and CSS in Depth

Leadership and Extracurricular Activities

- **Editor**, Computer Engineering Society (CENSOC), Faculty of Engineering, USJ 2025
- **Organizing Committee Member**, NerdCircle 24', Computer Engineering Society, USJ 2024
- **Hackathon Participant**, InnovateX Hackathon 2025, Zebra Technologies October 2025
- **Student Member**, Institution of Engineers Sri Lanka (IESL)
- **Member**, Rotaract Club, University of Sri Jayewardenepura 2022 – Present

Languages

Sinhala: Native

English: Fluent

References

Dr. Udaya Wijenayake

Head of the Department, Department of Computer Engineering

Faculty of Engineering, University of Sri Jayewardenepura

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